

Homework Assignment 2

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Fundamentals of Data Management - Module 2

Step 1: Import the Data

First we load the `readxl` package, which lets R read Excel files. Then we import Sheet 3 of our spreadsheet, which contains the 5 fake study participants.

```
# Load the package that reads Excel files
library(readxl)

# Import the fake data from Sheet 3 of our Excel file
# Make sure Gonzalez_hw2.xlsx is in the same folder as this .Rmd file
data <- read_excel("Gonzalez_hw2.xlsx", sheet = "Fake_Data_5_Participants")

# Print the full dataset so we can confirm it loaded correctly
print(data)
```



```
## # A tibble: 5 x 20
##   respondent_num survey_date respondent_initials zip_code date_of_birth sex
##           <dbl> <chr>           <chr>           <chr>    <chr>      <dbl>
## 1             1 05/01/2026   A.G.             10001    03/15/1990    1
## 2             2 05/02/2026   B.R.             90210    07/22/1985    2
## 3             3 05/03/2026   C.M.             77001    11/05/2000    1
## 4             4 05/04/2026   D.T.             60601    01/30/1975    2
## 5             5 05/05/2026   E.L.             30301    09/14/2010    1
## # i 14 more variables: hh_relationship <dbl>, height_cm <dbl>, weight_kg <dbl>,
## #   bp_systolic <dbl>, bp_diastolic <dbl>, ever_attended_school <dbl>,
## #   years_of_school <dbl>, currently_in_school <dbl>, pet_dog <dbl>,
## #   pet_camel <dbl>, pet_hamster <dbl>, pet_rabbit <dbl>, pet_skunk <dbl>,
## #   survey_notes <chr>
```

Step 2: Summary Statistics for Height and Weight

Here we calculate the mean, median, minimum, and maximum for both `height_cm` and `weight_kg`.

```

# HEIGHT (cm)
cat("=== Height (cm) Statistics ===\n")

## === Height (cm) Statistics ===

cat("Mean:  ", mean(data$height_cm),  "\n")

## Mean:    164

cat("Median:", median(data$height_cm), "\n")

## Median: 165

cat("Min:   ", min(data$height_cm),   "\n")

## Min:    140

cat("Max:   ", max(data$height_cm),   "\n")

## Max:    182

cat("\n")

# WEIGHT (kg)
cat("=== Weight (kg) Statistics ===\n")

## === Weight (kg) Statistics ===

cat("Mean:  ", mean(data$weight_kg),  "\n")

## Mean:   64.4

cat("Median:", median(data$weight_kg), "\n")

## Median: 62

cat("Min:   ", min(data$weight_kg),   "\n")

## Min:    35

cat("Max:   ", max(data$weight_kg),   "\n")

## Max:    90

```

Step 3: Print Rows 2 and 3

Here we print only rows 2 and 3 from the dataset.

```
# Print rows 2 and 3 (all columns)
cat("=== Rows 2 and 3 ===\n")
```

```
## === Rows 2 and 3 ===
```

```
print(data[2:3, ])
```

```
## # A tibble: 2 x 20
##   respondent_num survey_date respondent_initials zip_code date_of_birth sex
##           <dbl> <chr>           <chr>           <chr>   <chr>         <dbl>
## 1             2 05/02/2026 B.R.             90210   07/22/1985     2
## 2             3 05/03/2026 C.M.             77001   11/05/2000     1
## # i 14 more variables: hh_relationship <dbl>, height_cm <dbl>, weight_kg <dbl>,
## #   bp_systolic <dbl>, bp_diastolic <dbl>, ever_attended_school <dbl>,
## #   years_of_school <dbl>, currently_in_school <dbl>, pet_dog <dbl>,
## #   pet_camel <dbl>, pet_hamster <dbl>, pet_rabbit <dbl>, pet_skunk <dbl>,
## #   survey_notes <chr>
```